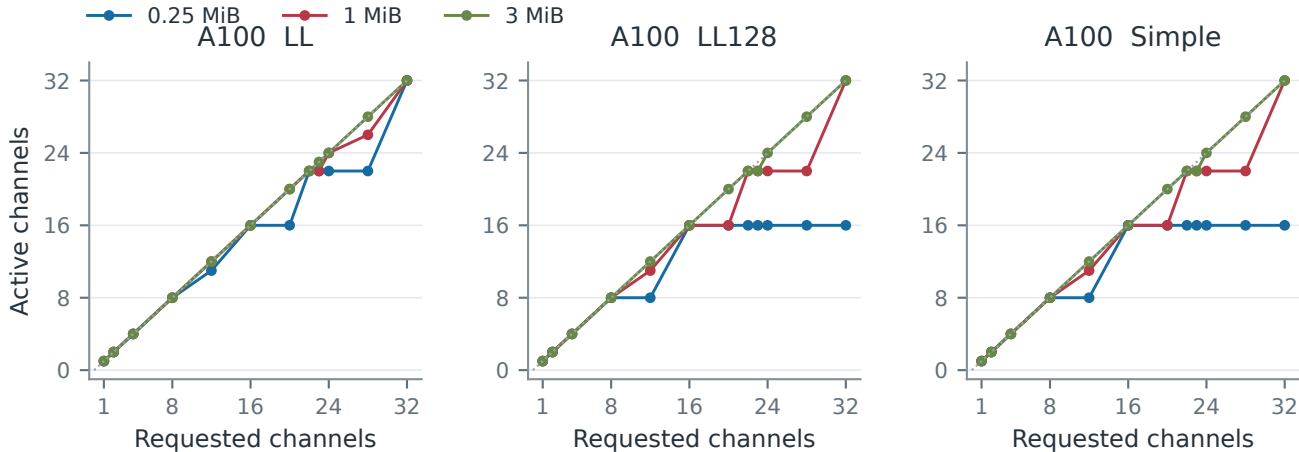


# A channel request is a budget, not the executed channel count

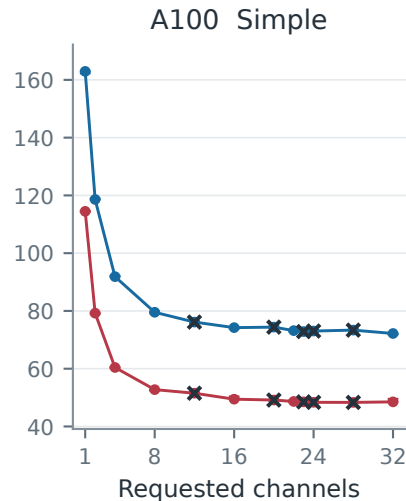
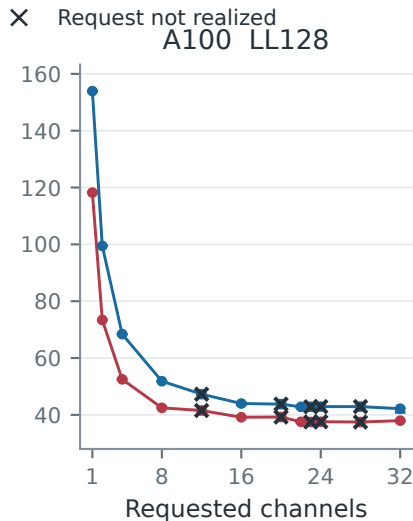
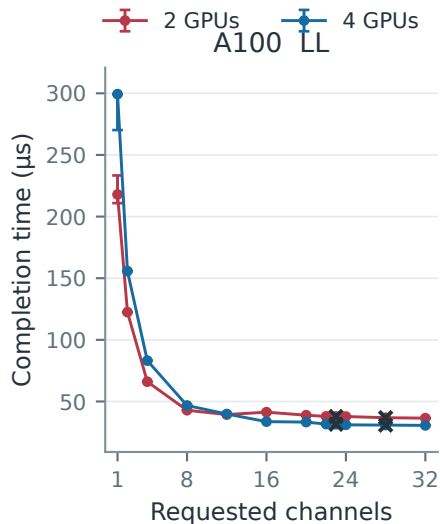
Separate selection observers; four GPUs; no timing values used



NCCL partitions the payload into work cells. Fewer active channels invalidate a fixed-request causal contrast.

# Channel count changes overlap and shared work

Ordinary GPU-event timing; 1 MiB per GPU; medians and interquartile ranges

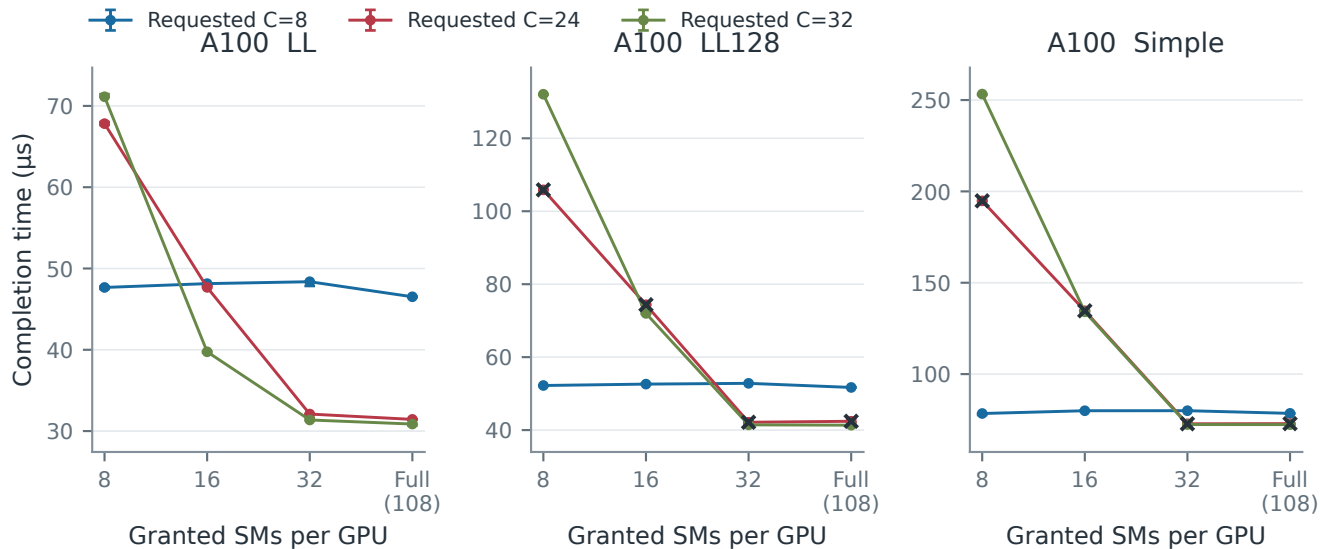


Five independent processes per point. Crosses retain observations whose requested control was not realized.

Identification data only. Each allocation is also summarized separately; no model curve or fitted band is shown.

# Restricting GPU execution leaves the link hardware unchanged

Four GPUs; 1 MiB per GPU; source-valid protocol and verified stream context

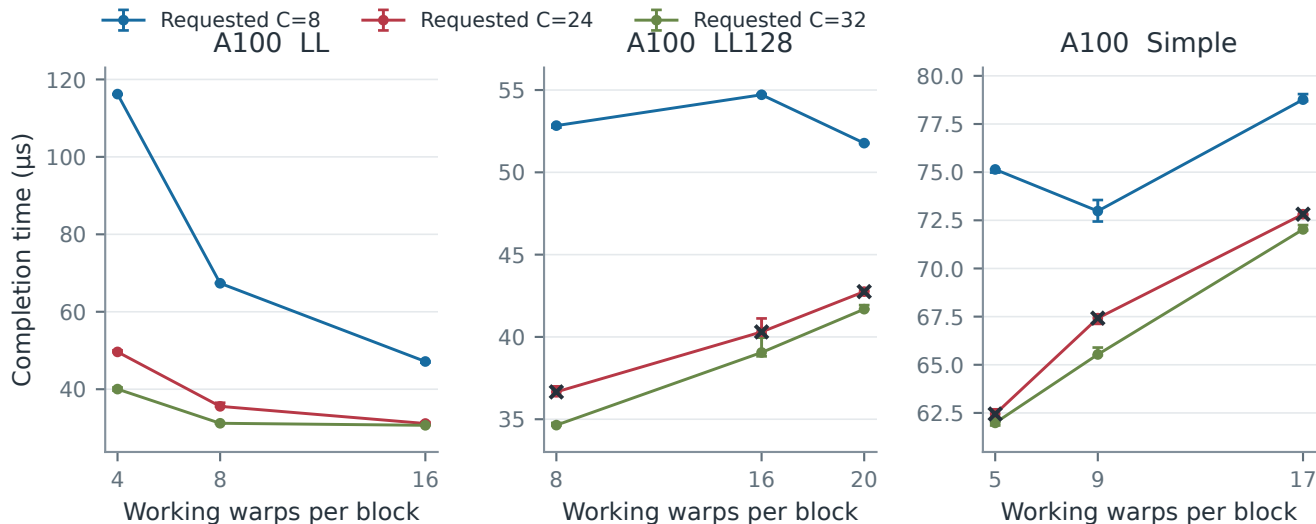


8/16/32 SMs use CUDA green contexts; Full uses an ordinary stream. Bars show five-process interquartile ranges.

SM: streaming multiprocessor. Crosses mark unrealized channel requests; no pure SM-cost attribution is implied.

# Warp count changes work overlap and block footprint

Four GPUs; 1 MiB per GPU; actual working warps from the separate observer

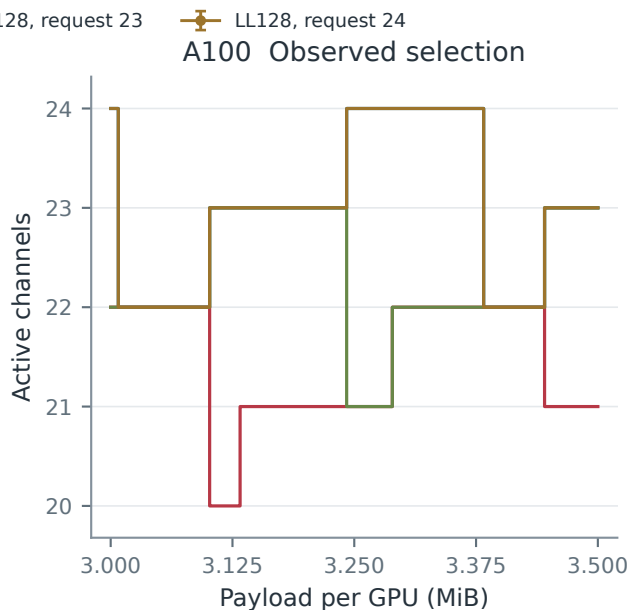
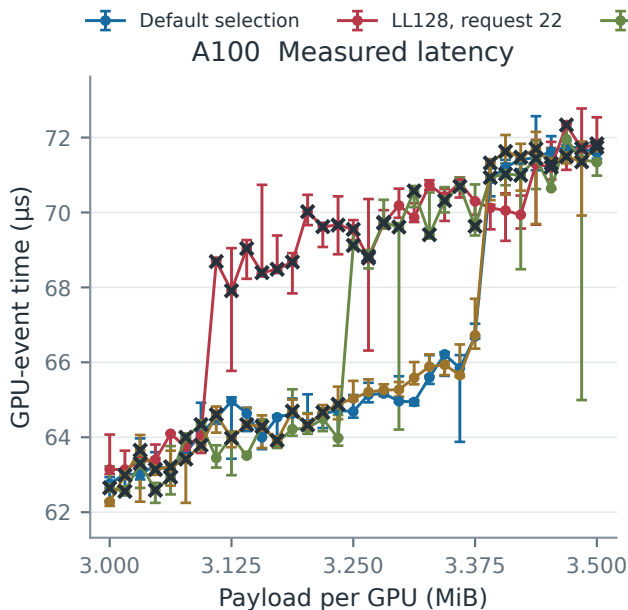


One warp contains 32 threads. Simple includes its additional synchronization warp. Bars show process IQRs.

Crosses mark unrealized channel requests. More warps are not assumed to be faster or to imply more SMs.

# A denser look at the four-GPU residual window

33 payloads at 16-KiB spacing; five ordinary timing processes per point

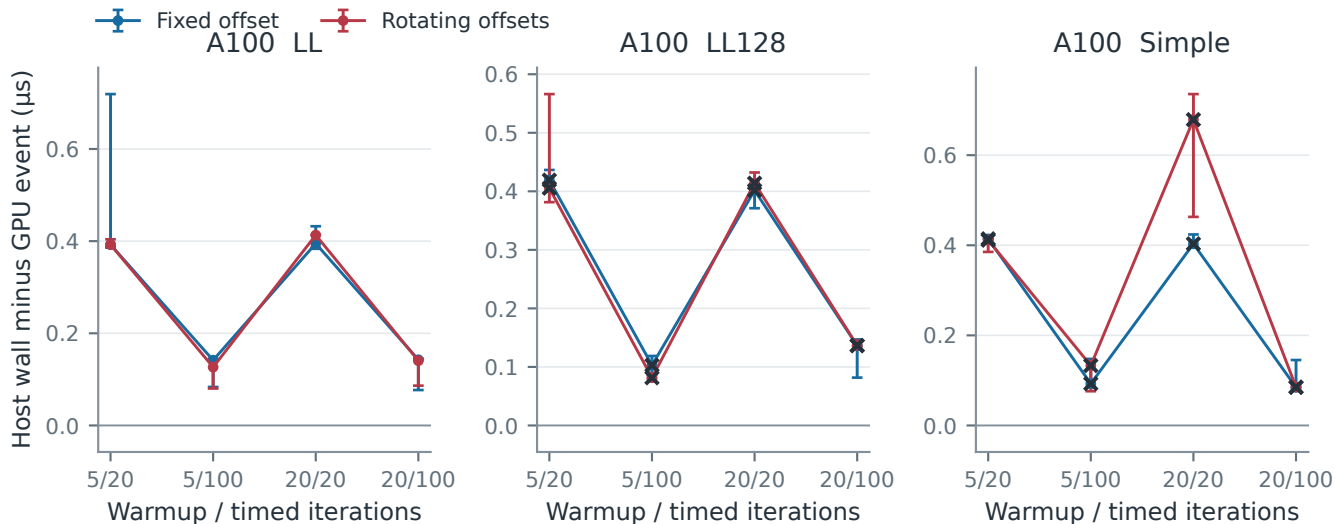


Bars are process interquartile ranges. Crosses mark unrealized requested channels, retained as observations.

Source work and actual channel partitions must explain these curves before any new hardware accuracy claim.

# Timer boundaries are paired on the same invocation

Four GPUs; 1 MiB per GPU; five independent processes per condition



Requested C=24. Crosses mark unrealized channel requests; paired timer differences remain descriptive.

Both timers include their own boundaries around the same launch loop. GPU events may include host issue gaps.

A paired difference is a measurement-boundary effect; it is not an isolated kernel-entry or host-service constant.